Mahalanobis Distance based Client Reliability Measurement in Federated Learning

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Introduction

Federated Learning (FL) enables distributed machine learning across devices or servers that hold localized data, eliminating the need to centralize this data due to stringent policy or privacy considerations. FL is typically orchestrated by a central server that coordinates with multiple client devices, each containing a subset of the overall data. The fundamental algorithm for Federated Learning operates in the following manner:

- **Model Initialization:** The central server initializes a global model and distributes it to all participating clients.
- **Local Training:** Each client trains the model locally using its private data for a set number of epochs, thus creating locally updated models.
- **Model Aggregation:** The updated models are sent back to the central server, which aggregates them (commonly using weighted averaging) to produce a new global model.
- **Global Model Update:** The updated global model is then shared with the clients for further rounds of training, repeating the process until convergence or until a specified number of iterations is reached.

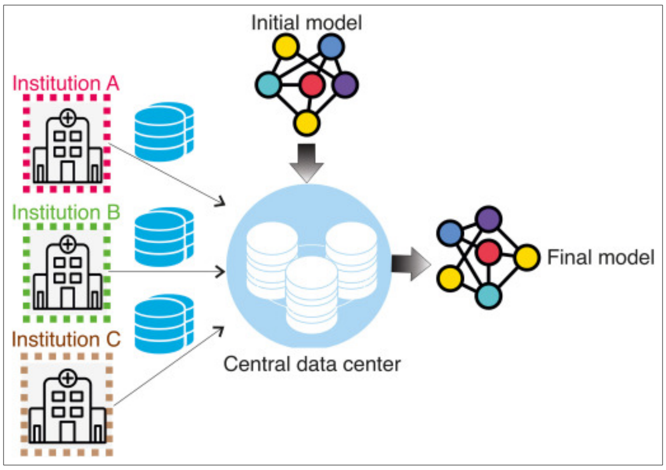

Figure 1: Federated Learning in the medical setting

The performance of the global model is highly sensitive to the quality of client gradient updates. Even a small proportion of unreliable clients can significantly degrade the model's accuracy and prevent it from converging. This is catastrophic in scenarios such as medical AI where the model's reliability is paramount. Unreliable clients can be broadly classified into two categories:

- **Malicious Clients:** These clients intentionally introduce harmful updates to degrade the global model. Typical attacks include:
  - **Sign Flip Attack:** Flip the signs of gradient updates
  - **Additive Noise Attack:** Add noise to the gradient updates
  - **Label Flip Attack:** Flip the labels of the training data to train inaccurate mdoels
- **Clients trained on unfavorable data:** These clients possess data with inherent imperfections due to human or machine errors. Examples include image artifacts such as blurring, distortions, blood cells, etc.

The key assumption that allows us to tackle this problem is that bad client updates will have a very different distribution when compared to benign updates. This is especially useful because it is observed that only such updates lead to degradation in performance, and even 'bad' updates which have a similar distribution to good updates help improve model performance by increasing its capability to generalize. Therefore, research in this area is focused on devloping measures that can detect and quantify this distribution shift, thus allowing us to separate benign and unreliable client updates and exclude the latter from the final model aggregation.

Previous Work

Most distance based algorithms to detect unreliable clients in federated learning use either cosine similarity or euclidean distance as their discriminator. Given below are some of the key works in this area:

- **Clustered Federated Learning:** Cosine similarity is used to form clusters of clients which aggregate updates within them.
- **Krum/M-Krum:** Uses euclidean distance to find the most reliable M updates which are then aggregated
- **ASMR:** Uses pairwise cosine similarity to detect unreliable client updates.

These measures are either applied on the flattened vector of the entire model or, more effectively, on the individual layers of the deep neural network (after which the results are aggregated using either a maxima/minima or average function).

Our Measure: Mahalanobis Distance

Mahalanobis Distance is a statistical method used to determine the distance between a point and a distribution. Mathematically, it is given as:

$$D_M(x) = \sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)} \tag{1}$$

where:

- x is the point whose distance is to be calculated
- μ is the mean of the distribution
- Σ is the covariance matrix of the distribution

Unlike traditional measures such as Euclidean distance, which treat each feature independently, Mahalanobis Distance accounts for correlations among features, thus providing a more robust way to evaluate how an observation deviates from a distribution. This is also a much more intuitive measure for our problem, since we are primarily trying to detect gradient updates which deviate significantly from the overall distribution and thus would have a high mahalanobis distance. Further, Lee et al. (2018) established that Mahalanobis distance is a very effective out-of-distribution detector in traditional machine learning pipelines *(Kimin Lee, Kibok Lee, Honglak Lee, Jinwoo Shin, "A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks", arXiv preprint arXiv:1807.03888v2, 2018).*

The primary challenge in computing the Mahalanobis distance for gradient updates lies in the requirement for an invertible covariance matrix, which necessitates that the number of samples exceeds the number of features. This poses a significant difficulty because deep learning models often contain millions of parameters, while the number of clients is typically limited (around 10-100 in most experimental settings). As a result, it becomes infeasible to directly apply the Mahalanobis distance except in specific scenarios. One such specific example is its application to convolutional filters within gradient updates. For instance, a convolutional layer with dimensions $256 \times 32 \times 32$ can be separated into 256 convolutional filters, each with dimensions of 32×32. When combining data from 10 clients, this yields 2560 filters, each containing 1024 parameters, allowing for an invertible covariance matrix to be computed and enabling the use of Mahalanobis distance. Promising results were obtained on using just the convolutional layers from models to detect unreliable updates, which motivated us further to find a more general method which can be used on any layer

The final method developed leverages the mathematical equivalence between the Mahalanobis distance and Euclidean distance when data is transformed into its Principal Component Analysis (PCA) space. More specifically, we perform the following steps, which are collectively known as 'whitening' of the data:

- Standardization of data to achieve zero mean and unit variance
- Principal Component Analysis

In the transformed space, the L-2 Euclidean norm corresponds to the Mahalanobis distance in the original space (a formal proof of this equivalence can be found in Appendix A). This approach provides the exact value of the Mahalanobis distance when there are sufficient samples, and offers a close approximation when sample sizes are limited. Furthermore, its reliability is demonstrated by the observation that, for most layers, the bulk of the variance is captured within the first 3-4 principal components. This ensures that we retain most of the critical variance without any substantial loss of important information.

Experiments

Datasets

Two medical image datasets were used to evaluate the performance of the proposed method:

- **NCT-CRC-HE:**
  - 100k slide images of human colorectal cancer and normal tissue- 9 classes of tissue types
  - All images originate from the same hospital and similar equipment, which ensures that they are **independent and identically distributed**.

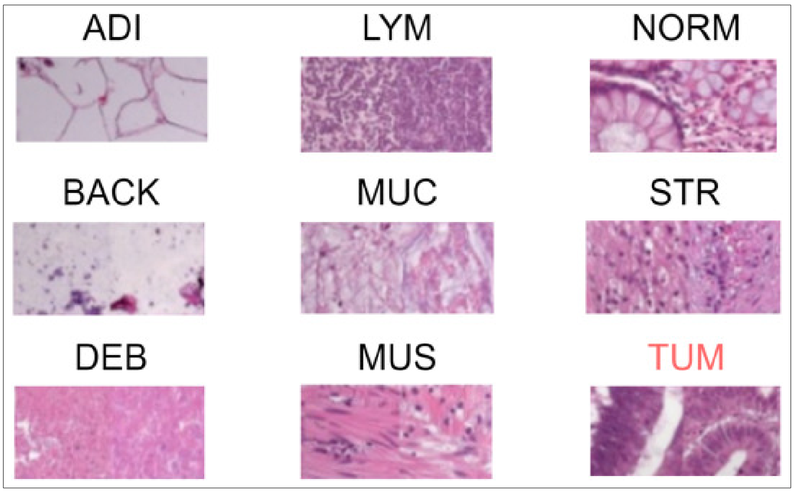

Figure 2: Images from NCT-CRC-HE dataset

- **Camelyon-17:**
  - Binary Classification problem- Whether slide has tumor tissue or not
  - The data is sourced from five different hospitals, each with potentially varying equipment and imaging conditions, resulting in **non-independent and non-identically distributed (non-IID) data**.

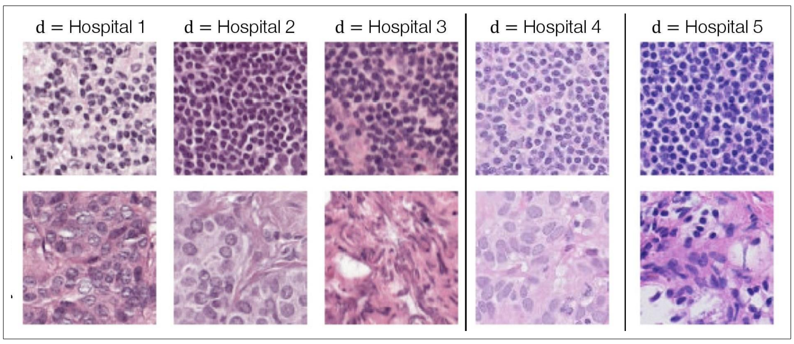

Figure 3: Images from Camelyon-17 dataset

Training setup

To evaluate the proposed method, we simulated a federated learning setup with the following parameters:

- **Number of Clients Simulated**: 10 clients were simulated (8 training + 2 testing).

- **Global and Local Epochs**: 10 overall global epochs, 1 local epoch per round for each client
- **Model Architecture**: Untrained DenseNet-121 model
- **Local Optimization**: Adam optimizer with initial learning rate 0.01 was used for local training
- **Aggregation Algorithm**: FedAvg (Federated Averaging) algorithm

Out of the 8 training clients, 2 were designated as unreliable clients. The unreliable clients simulated the following types of client malfunctionings:

- **Sign Flip Attack**: Clients flip the signs of gradient updates
- **Additive Noise Attack**: Gaussian noise with zero mean was introduced at varying levels, with variances set to 10%, 25%, 50%, and 75% of the original variance of the gradient updates respectively
- **Image Artifacts**: The FrOoDo framework was used to introduce various image artifacts, including dark spots, grease spots, squamos tissue, threads, blood artifacts, and bubbles, to test the effect of diverse data corruption scenarios on model updates.

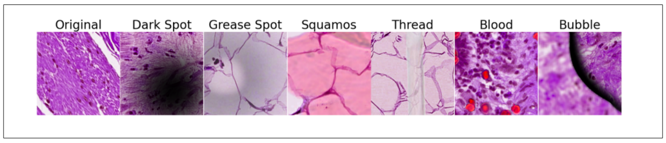

Figure 4: Image Artifacts in cell images using the FrOoDo framework

The Mahalanobis distance was computed separately for each layer of a client's update, and then an average across all layers was calculated to produce a single final distance measure for each client. Pairwise euclidean distance and cosine similarity were also computed for comparison.

Results

Cosine similarity, Euclidean distance, and Mahalanobis distance operate on different scales, making direct comparisons impractical. Instead, we assess the relative scaling effect on each client when Mahalanobis distance is used in place of Euclidean distance or cosine similarity. Specifically, if the Mahalanobis-to-Euclidean ratio is represented as a for normal clients and b for malfunctioning clients, we report the ratio b/a. **A higher value of this ratio indicates greater scaling for malfunctioning clients compared to normal clients (which means that malfunctioning clients are farther from the distribution than normal clients), thus demonstrating superior performance of Mahalanobis distance relative to the other measures.**

Note: Certain values are missing from the graphs below due to either numerical instability in calculating the distance measures or the model's inability to converge under conditions of severe client malfunctioning.

The results indicate that a majority of layers exhibited improvements across all malfunctioning settings, demonstrating that Mahalanobis distance can effectively serve as a strong alternative to traditional measures.

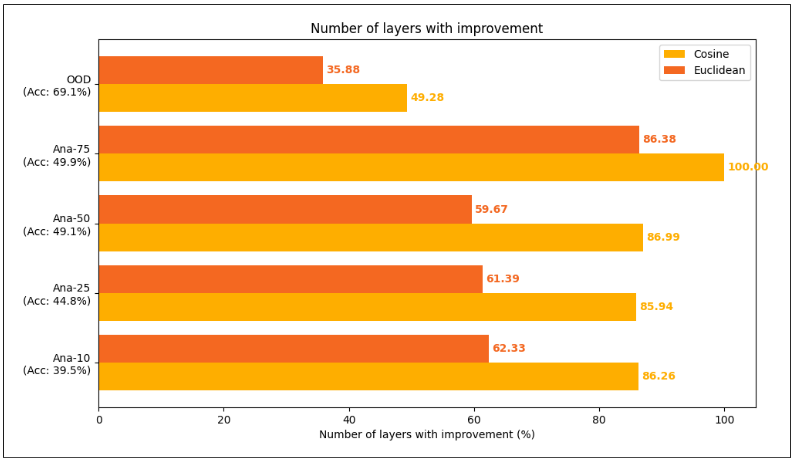

Figure 5: Percenrtage of Layers that show improvement for Camelyon17 dataset

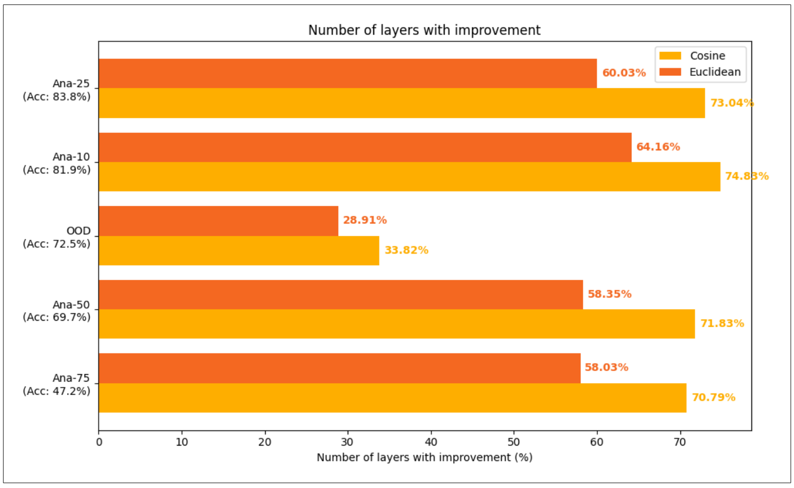

Figure 6: Percenrtage of Layers that show improvement for NCT-CRC-HE dataset

The most consistent improvements were observed in three specific types of layers: the convolutional layers, the running mean layers, and the running variance layers. This is particularly noteworthy, as algorithms can be fine-tuned to concentrate on these layers for detecting unreliable clients.

The corresponding results for Camelyon-17 are presented below:

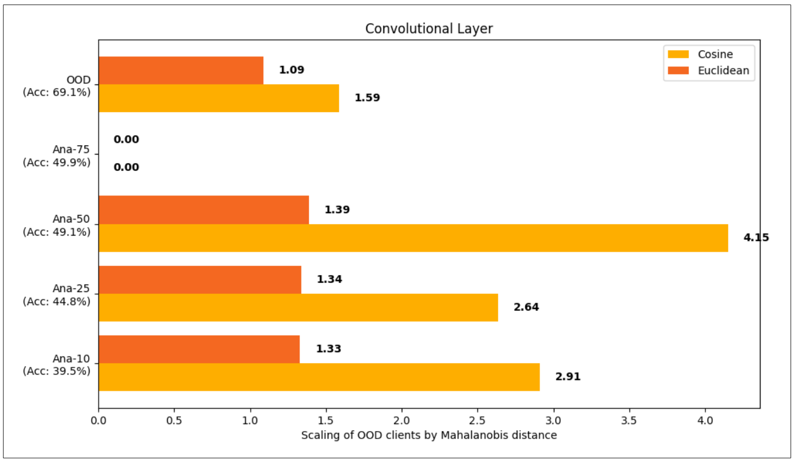

Figure 7: Scaling of convolutional layer for Camelyon17 dataset

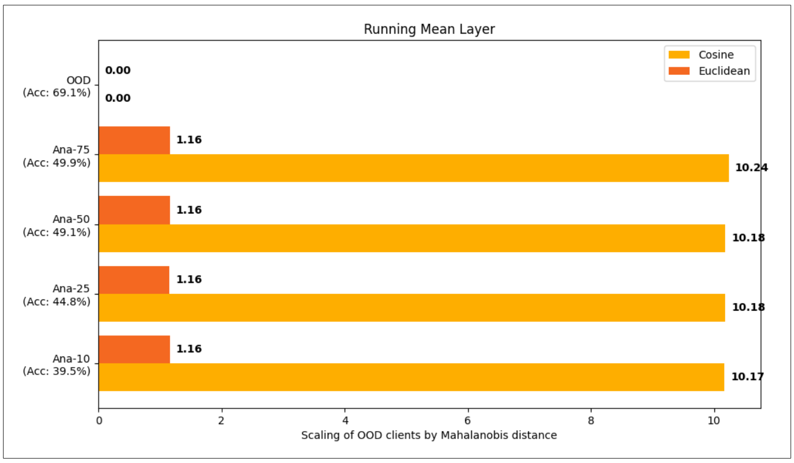

Figure 8: Scaling of running mean layer for Camelyon17 dataset

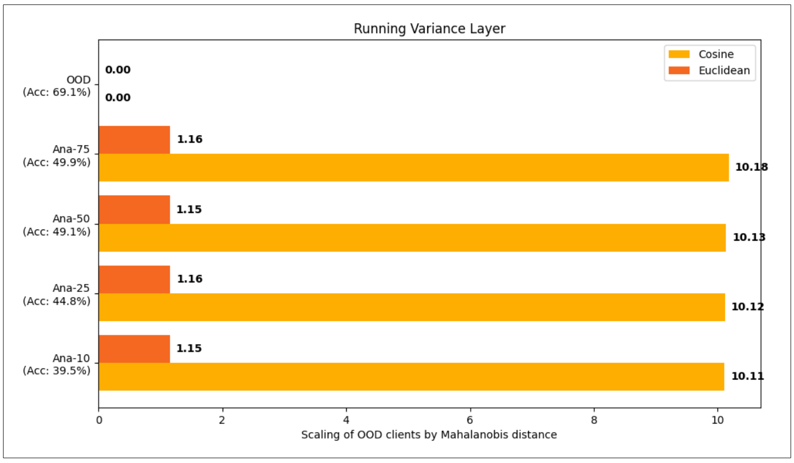

Figure 9: Scaling of running mean layer for Camelyon17 dataset

A similar boost is seen for the NCT-CRC-HE dataset.

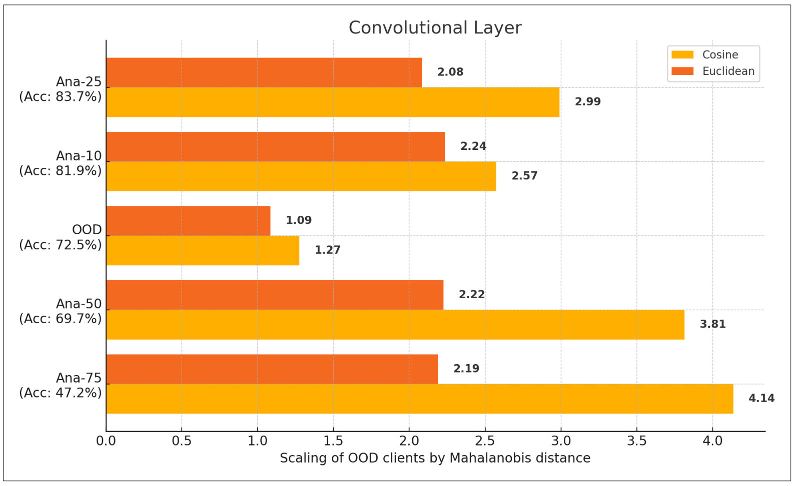

Figure 10: Scaling of convolutional layer for NCT-CRC-HE dataset

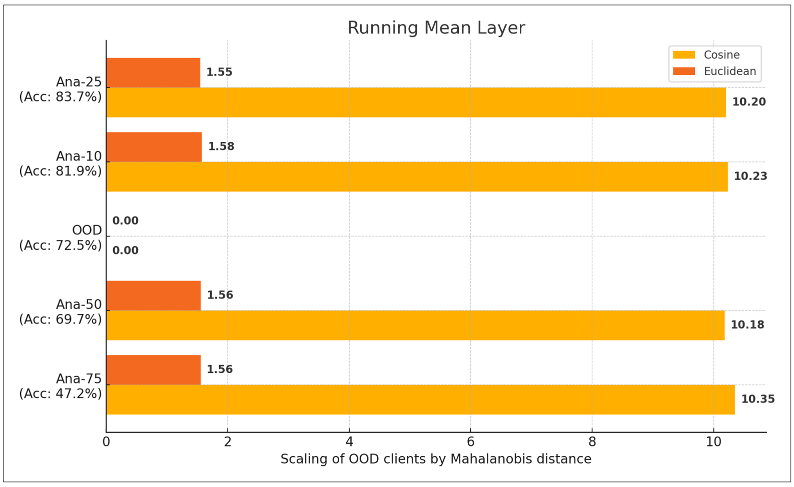

Figure 11: Scaling of running mean layer for NCT-CRC-HE dataset

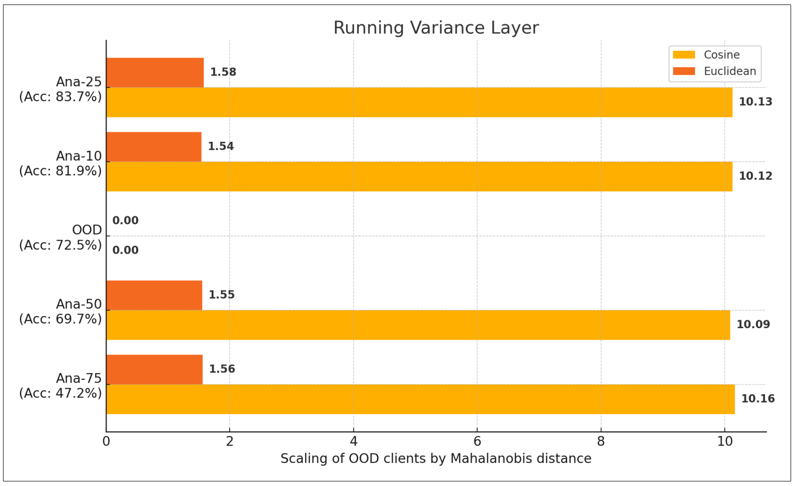

Figure 12: Scaling of running variance layer for NCT-CRC-HE dataset

Overall, Mahalanobis distance significantly outperforms cosine similarity and, in most settings, surpasses Euclidean distance, with performance at least matching that of Euclidean distance in the worst-case scenarios, as evidenced by the results below.

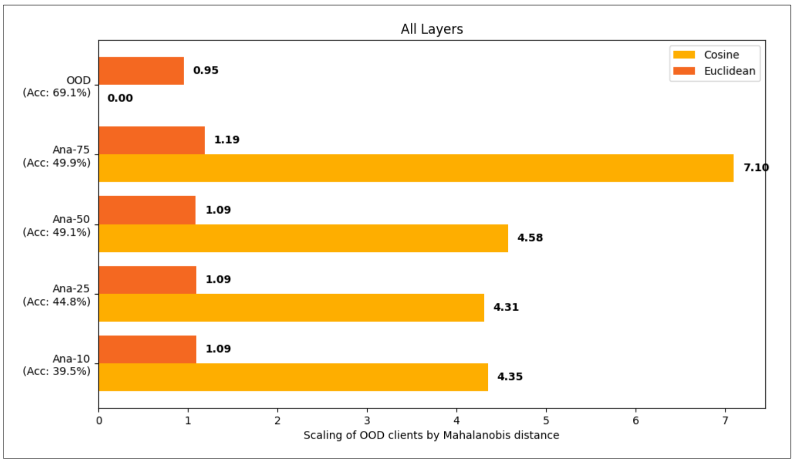

Figure 13: Average performance over all layers for Camelyon17 dataset

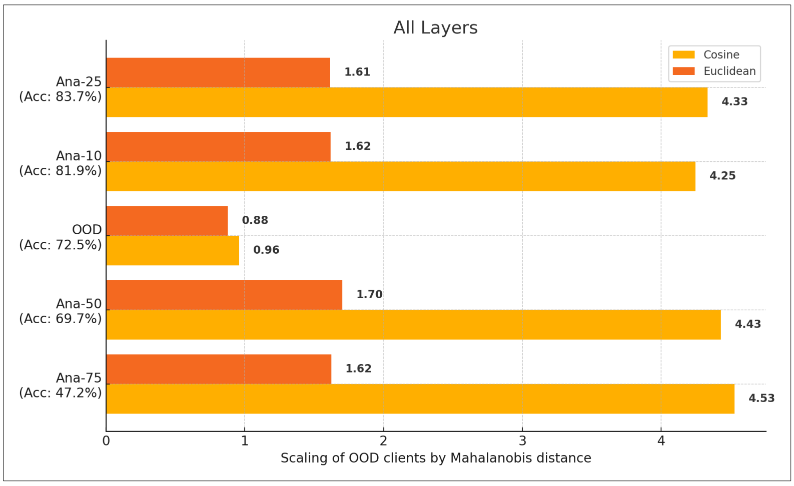

Figure 14: Average performance over all layers for NCT-CRC-HE dataset

Conclusion

The findings demonstrate that Mahalanobis distance is a strong alternative to traditional measures such as cosine similarity and Euclidean distance to detect malfunctioning clients in Federated Learning. By measuring the distance from a distribution rather than simple point-wise comparisons, Mahalanobis distance is more intuitively justifiable and robust. Notably, the improvements are concentrated in specific layers, suggesting potential avenues for fine-tuning and optimizing algorithms for targeted client reliability measurement. The observed improvements across a range of malfunctioning settings validate the effectiveness of Mahalanobis distance in practical scenarios. Future work can focus on replacing existing distance measures in established algorithms with Mahalanobis distance to further evaluate its performance across various federated learning applications.

Appendix A: Proof of Equivalence between Mahalanobis distance and Euclidean Norm in PCA-whitened space

Mahalanobis distance is defined as

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \mu)^\top \mathbf{\Sigma}^{-1} (\mathbf{x} - \mu)}$$

Now, consider the eigen decomposition of the covariance matrix $\mathbf{\Sigma}$ as

$$\mathbf{\Sigma} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$$

Here:

- $\mathbf{Q}$ is an orthogonal matrix whose columns are the eigenvectors of $\mathbf{\Sigma}$.
- $\mathbf{\Lambda}$ is a diagonal matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the diagonal.

Substituting this in the equation for Mahlanaobis distance, we get,

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \mu)^\top \mathbf{Q}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top (\mathbf{x} - \mu)}$$

Now, let $\mathbf{y} = \mathbf{Q}^\top (\mathbf{x} - \mu)$. This is the PCA transformation where the original data $\mathbf{x}$ is projected onto the principal components defined by $\mathbf{Q}$. The Mahalanobis distance now becomes:

$$D_M(\mathbf{x}) = \sqrt{\mathbf{y}^\top \mathbf{\Lambda}^{-1} \mathbf{y}}$$

To standardize the variance to 1, we scale the transformed coordinates $\mathbf{y}$ by the square root of the eigenvalues,

$$\mathbf{z} = \mathbf{\Lambda}^{-1/2} \mathbf{y}$$

where $\mathbf{\Lambda}^{-1/2}$ is the diagonal matrix with $1/\sqrt{\lambda_1}, 1/\sqrt{\lambda_2}, \ldots, 1/\sqrt{\lambda_n}$ on the diagonal. These two successive transformations achieve the whitening of the data. The Mahalanobis distance therefore becomes,

$$D_M(\mathbf{x}) = \sqrt{\mathbf{z}^\top \mathbf{z}} = \|\mathbf{z}\|_2$$

which is the euclidean norm of the transformed data $\mathbf{z}$.

Therefore, the Mahalanobis distance in the original space is equivalent to the Euclidean norm in the PCA-whitened space.